Nijat Abbasov

Last updated in Oct 2026

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Nijat Abbasov NijatAbbasov06

Education

KAIST - Korea Advanced Institute of Science and Technology *Sep 2023 - Present*

BS in Electrical Engineering and Mathematics double major, graduating in 2028.

- Currently a Junior student with a GPA: 3.77/4.3.
- **Coursework:** Robotics, Linear Systems, Control Systems, Signals and Systems, Probability and Stochastic Processes, Electromagnetics I & II

Experience

Huawei HKRC - Huawei Hong Kong Research Center *Hong Kong, China*

Research Intern *Huawei HKRC* *2025 Oct - 2026 June*

- Conducted research on state-of-the-art Deep Reinforcement Learning (DRL) algorithms for solving peg in hole problem for physical robotic manipulator. [Gymnasium, PyBullet, PyTorch]
- Utilized visual servo, tracking and grasping algorithms to build grasping pipeline for real robotic arm.
- Researched test-time and train-time scaling of LLM models that solve Physics Olympiad problems [PyMuPDF, PyTorch, smolagents]

ACSS - Autonomous Control for Stochastic Systems Undergraduate Researcher *Daejeon, South Korea*

Fault-Tolerant Routing in Dynamic Environments *ACSS* *2023 Dec - 2025 Oct*

- Simulated and compared Extended Kalman Filter and Kalman Filter implementation for rotors with non linear dynamics. [MATLAB, Simulink]
- Implemented a Multiplicative Extended Kalman Filter (MEKF) - a variant of EKF to estimate wind velocity with and without using an airspeed sensor in simulation. [Gazebo, ROS2, ROS, PX4]

Projects

GNSS-Denied UAV Geo-Localization with VIO Fusion *2026 Aug* *Github*

- Built a CPU-only system that bounds VIO drift with absolute fixes from satellite imagery, fusing OpenVINS (MSCKF) with self-supervised cross-view geo-localization (CAEVL).
- Rejected outliers with chi-square gating against the VIO prior and consensus checks, raising fix precision from 17.7% to 99%.
- Reduced position RMSE by 36% in real time (13.3 Hz on CPU) on 3.6 km of real flight data, and by 71% over the cruise segment offline. [OpenVINS, ROS2, Kalibr, PyTorch, Python]

Multi-Disciplinary Robotics *2025 Mar - 2025 June* *Github*

- Created a simulation environment for the autonomous navigation of a drone model in Gazebo.
- Utilized path searching algorithm in HKUST-Aerial-Robotics/Fast-Planner repository for quadcopter path searching.
- Implemented quadcopter localization with ORB-SLAM3 and AprilTags in simulation, and tested AprilTag detection in real-world experiments. [ROS, Gazebo, PX4]

Awards

TOPIK II - Test of Proficiency in Korean II Achieved level 3 - equivalent to B1 level in CEFR scale.

International Physics Olympiad 2023 Bronze medal

European Physics Olympiad 2022 Honorable Mention

European Physics Olympiad 2024 Conducted EuPhO Physics camp and participated as an observer in the olympiad.

International Astronomy and Astrophysics Olympiad 2022 Honorable Mention

National Physics Olympiad 3 Gold medals in consecutive years: 2021, 2022, 2023.

Skills

Programming Languages: C, C++, Python, Matlab, Bash, Latex, HTML, CSS.

Technologies: Simulink, ROS2, ROS, Gazebo, Px4, Arduino Uno, AutoDesk Fusion 360, PyTorch, Gymnasium, PyBullet

Languages: Turkish[C2], Azerbaijani[C2], English[C1], Korean[B2], Russian[B2], Chinese[A1]